$f(x) = b + \log_a x$ $\qquad f\left(\frac{1}{8}\right)$

$f(1) = -1$ $\qquad -1 = b + \underbrace{\log_a 1}_{=0}$

$f(2) = -3$

$b = -1$

$f(2) = -3$

$-3 = -1 + \log_a 2$

$-2 = \log_a 2$

$a^{-2} = 2$

$\frac{1}{a^2} = 2$

$a^2 = \frac{1}{2}$

$a = \sqrt{\frac{1}{2}}$

$f\left(\frac{1}{8}\right) = -1 + \log_{\sqrt{\frac{1}{2}}} \frac{1}{8} =$

$-1 + \frac{-3}{-1/2} \log_2 2 =$

$$-1+6=5$$

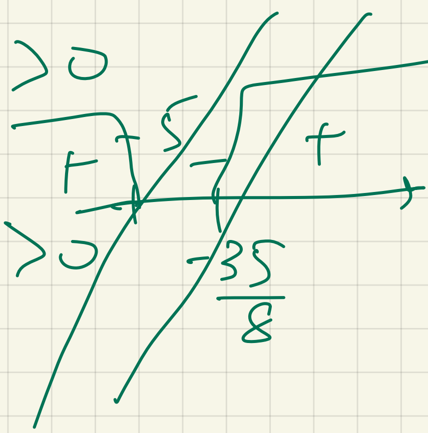
$$\log_{11}(8x^2+7)-\log_{11}(x^2+x+1)\geq \log_{11}\left(\frac{x}{x+5}+7\right)$$

$8x^2+7>0$ ✓

$x^2+x+1>0$ $\qquad -1\pm\sqrt{1-4}<0$

$\frac{x}{x+5}+7>0$ $\qquad \frac{x+7x+35}{x+5}>0$

$\frac{8x+35}{x+5}>0$



ОДЗ:

$x\in(-\infty;-5)\cup\left(-\frac{35}{8};+\infty\right)$

$$\frac{8x^2+7}{x^2+x+1}\geq\frac{x}{x+5}+7$$

$$\frac{8x^2+7}{x^2+x+1}\geq\frac{8x+35}{x+5}$$

$$(8x^2+7)(x+5)-(8x+35)(x^2+x+1)$$

$$y = a(x-3) \qquad x=3 \quad y=0$$

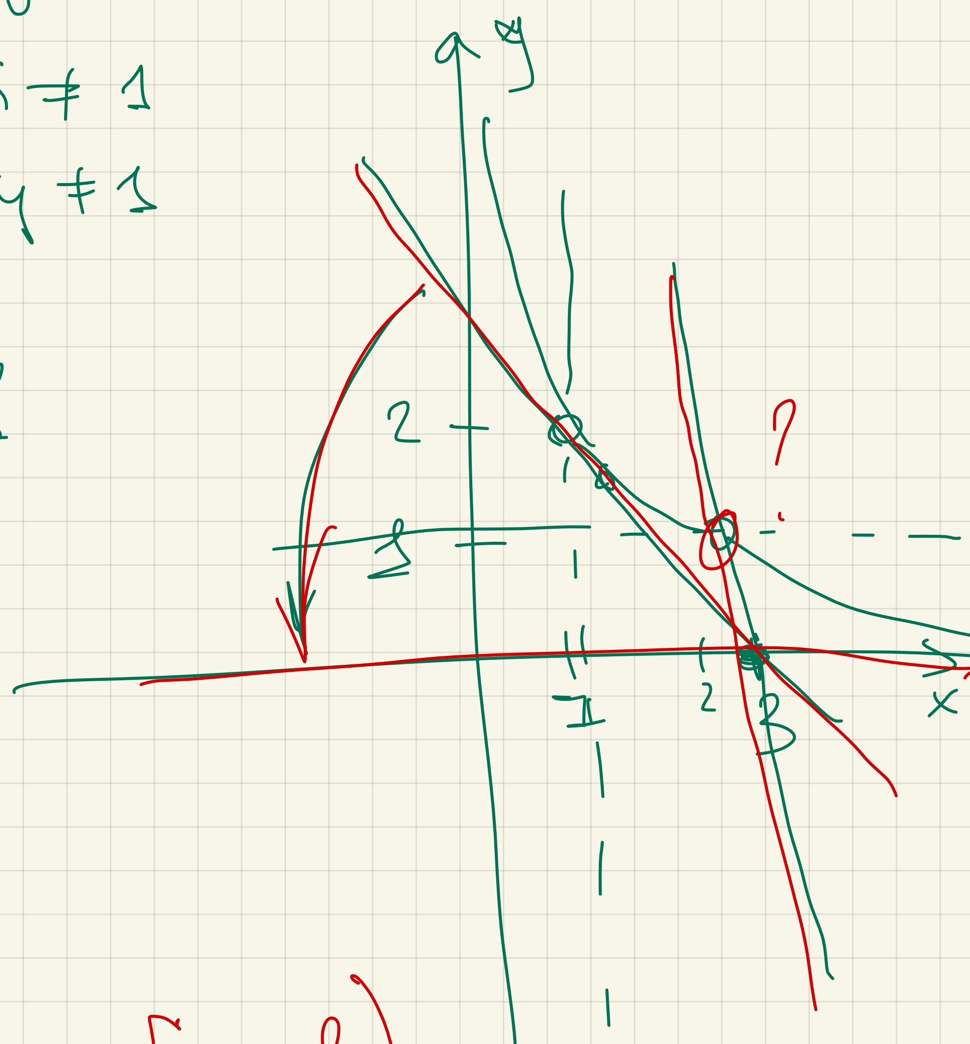
$$\frac{1}{\log_x 2} + \frac{1}{\log_y 2} = 1$$

$$\begin{cases} \log_2 x + \log_2 y = 1 \\ x>0 \qquad x \neq 1 \\ y>0 \qquad y\neq 1 \end{cases}$$

$$\begin{cases} \log_2 xy = 1 \\ x>0 \\ y>0 \end{cases}$$

$$xy = 2$$

$$y = \frac{2}{x}$$



$$a \in \left[0; -\frac{8}{9}\right) \cup \{-1\}$$

$$\frac{2}{x} = 6(x-3) \qquad \Delta = 0$$

$$\begin{cases} x^2 + 18px + 77p^2 \leq 0 \\ (x-324)^2 \geq (29p)^2 \end{cases}$$

$$\begin{cases} (x+7p)(x+11p) \leq 0 \\ (x-324-29p)(x-324+29p) \leq 0 \end{cases}$$

$$x^2 + \underset{2\cdot 9p\cdot x}{18px} + 81p^2 - 81p^2$$

$$(x+9p)^2 - 81p^2 + 77p^2$$

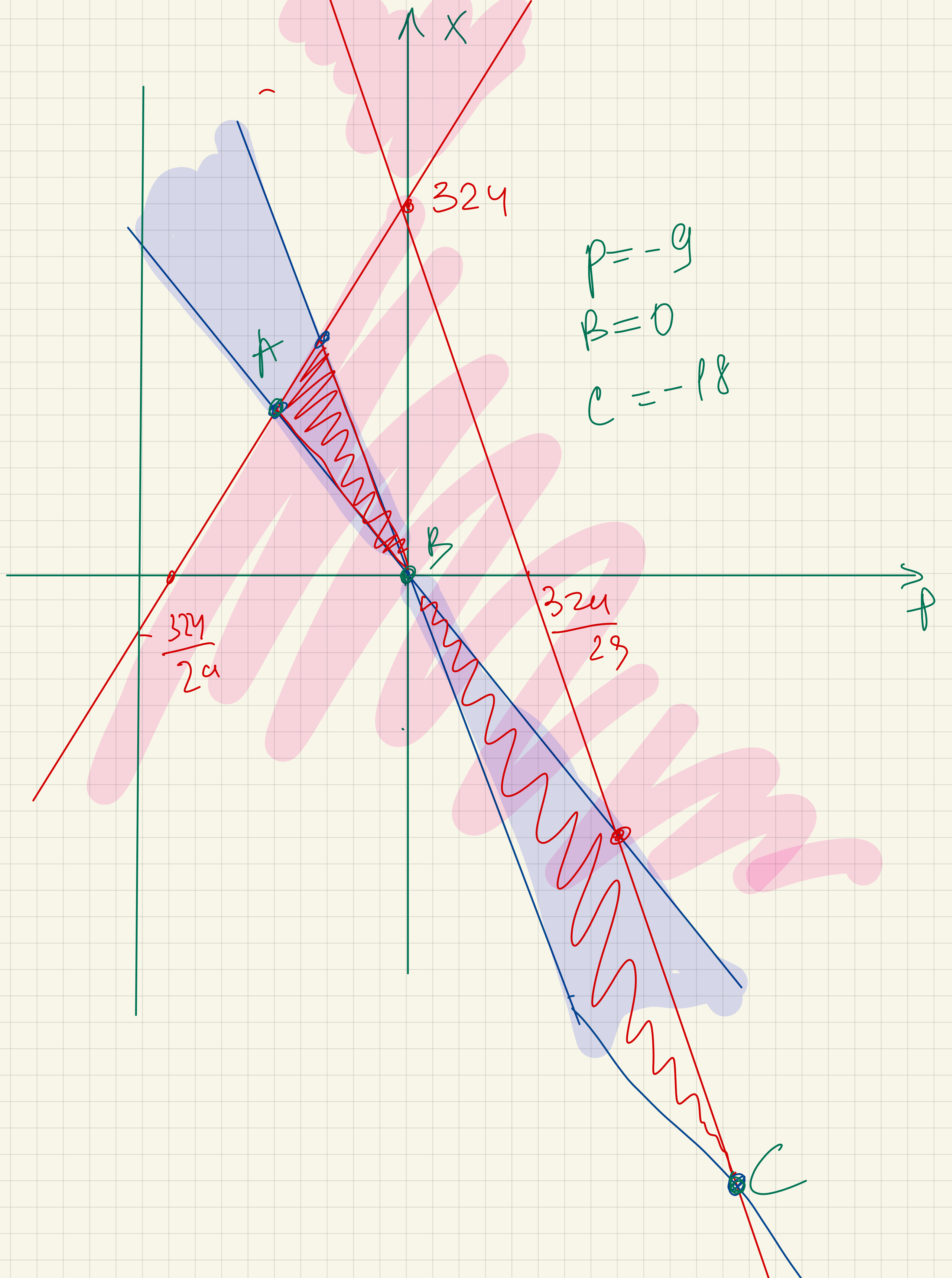
$$(x+9p)^2 - 4p^2 \leq 0$$

$$(x+9p-2p)(x+9p+2p) \leq 0$$

$$x = -7p$$

$$x = -11p$$

$$x = 29p + 324 \qquad x = -29p + 324$$



x
p
324
$\frac{324}{24}$
$-\frac{324}{24}$
A
B
C
p = -9
B = 0
C = -18

$$\begin{cases} \sqrt{x}\,(x^2-x+2)-yx^3=yx(2-x) \\ y^2+(2a-7)y+(a+2)(5-3a)=0 \end{cases}$$

2реш

$$\sqrt{x}(x^2-x+2)=yx^3+yx(2-x)$$

$$\sqrt{x}(x^2-x+2)=yx(x^2-x+2)$$

$$\begin{cases} (x^2-x+2)(\sqrt{x}-yx)=0 \\ y^2+(2a-7)y+(a+2)(5-3a)=0 \end{cases}$$

теор. $y_1\cdot y_2=(a+2)(5-3a)$

$y_1+y_2=7-2a$

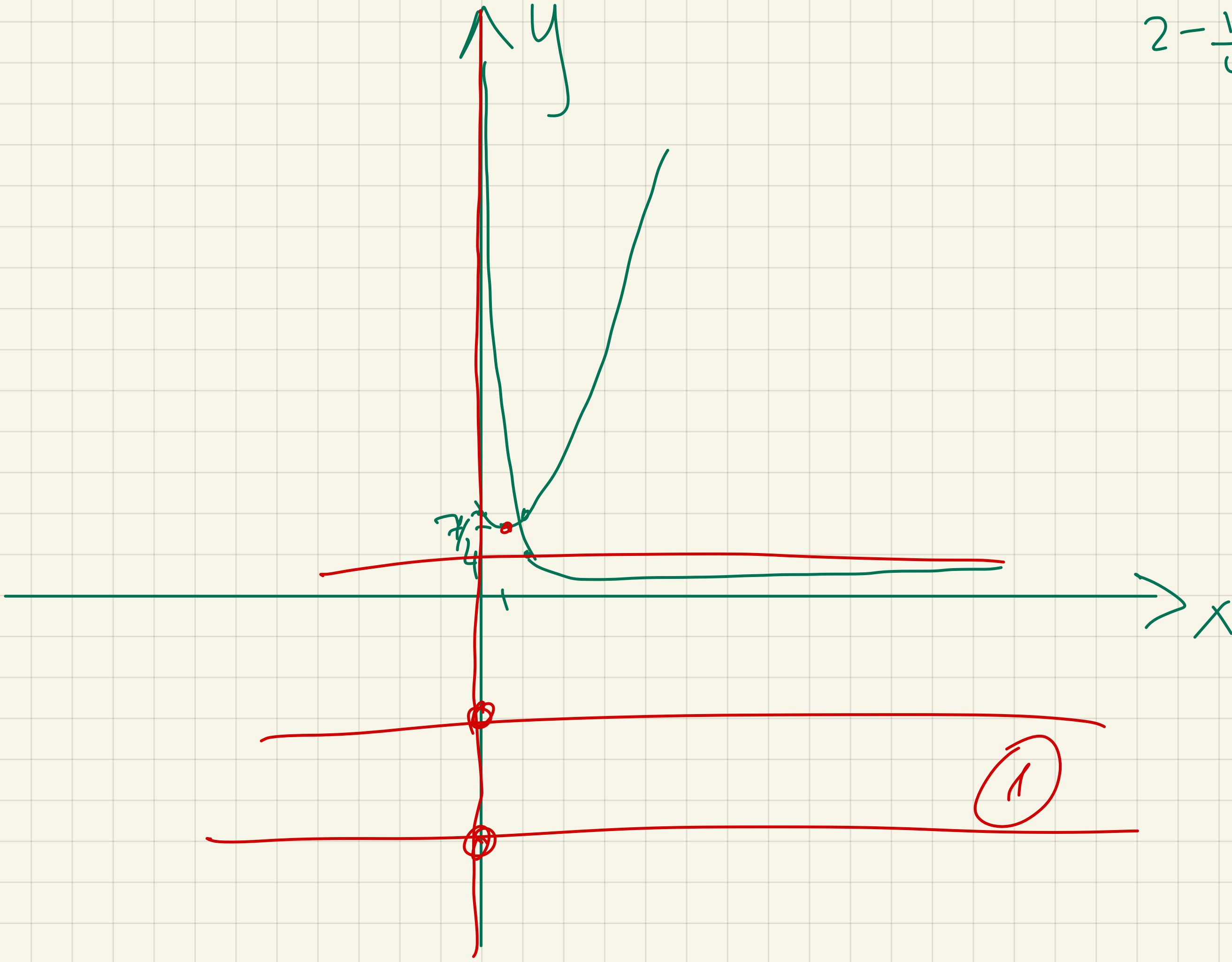
$$(y-a-2)(y-5+3a)=0$$

$\sqrt{x}=yx \qquad x\geqslant 0$

$x_0=\frac{1}{2}$

$y_0=\frac{1}{4}-\frac{1}{2}+2=$

$$2-\frac{1}{4}=\frac{7}{4}$$



$$y = a+2 \qquad y = -3a+5$$

$$\begin{cases} yx \geq 0 \\ x \geq 0 \\ y^2x^2 = x \ \checkmark \end{cases} \longrightarrow x(y^2x-1)$$

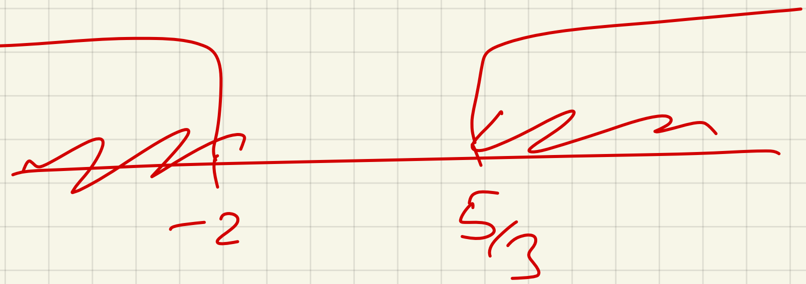
$$x = 0$$

$$x = \frac{1}{y^2}$$

$$y^2 = \frac{1}{x} \Rightarrow y = \pm\sqrt{\frac{1}{x}}$$

①

$$\begin{cases} a+2 \le 0 \\ -3a+5 \le 0 \end{cases}$$



$a \le -2$

$3a \geq 5$

$a \geq \frac{5}{3}$

$\varnothing$

②

$a+2 = -3a+5$

$4a = 3$

$a = \frac{3}{4}$

Ответ: $\frac{3}{4}$